SINGAPORE DIMENSIONS • GRADE 1A • CHAPTER 8

Making 10 to Add (Bridging Through 10)

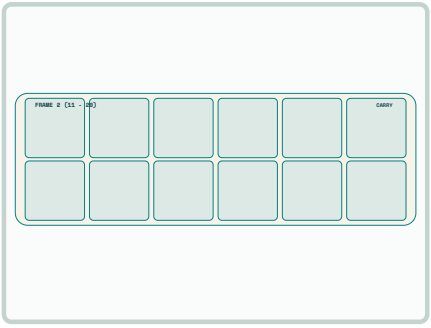
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.1.OA.C.6

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Ten Frame Shelf

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Frame A has 8 counters; Frame B has 5. Move 2 from B to A to make a full 10.

Sketch Model on Paper:
Draw 8 dots in Frame 1, 5 dots in Frame 2, arrow moving 2 dots.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE: 123 Base-10 Arithmetic

1.	9	X: 9	Key 1st operand
2.	ENTER	Y: 9 | X: 9	Stack lift copies X
3.	1	Y: 9 | X: 1	Key 2nd operand
4.	+	X: 15	Drop stack & compute

EXPECTED LCD X: 15 (10 + 5)

1 **Bridge through 10 to solve 8 + 5: 8 + 2 + 3.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Bridge through 10 to solve 7 + 6: 7 + 3 + 3.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Bridge through 10 to solve 9 + 7: 9 + 1 + 6.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Bridge through 10 to solve 8 + 7: 8 + 2 + 5.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 8

Making 10 to Add (Bridging Through 10) • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Ten-Frame Studio (Dual Frame Dock)

CAD Blueprint: ten_frame_shelf

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/lib/ten_frame_shelf.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Teal Dual Stacking Trays (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Bridge through 10 to solve 8 + 5: 8 + 2 + 3.** Standard: CCSS.MATH.1.OA.C.6	13	[8] [ENTER] [2] [+] [3] [+]
2	**Bridge through 10 to solve 7 + 6: 7 + 3 + 3.** Standard: CCSS.MATH.1.OA.C.6	13	[7] [ENTER] [3] [+] [3] [+]
3	**Bridge through 10 to solve 9 + 7: 9 + 1 + 6.** Standard: CCSS.MATH.1.OA.C.6	16	[9] [ENTER] [1] [+] [6] [+]
4	**Bridge through 10 to solve 8 + 7: 8 + 2 + 5.** Standard: CCSS.MATH.1.OA.C.6	15	[8] [ENTER] [2] [+] [5] [+]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Emphasize finding the complement of the first number to 10 first.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.